Lecture 9 // Sept 22 // Singularities in $\mathcal{M}$, part 2

Last time: We saw that connections whose stabilizer is larger than $\pm Id$ lead to singular points of $\mathcal{M}$, and this does happen.

If E is split and ∇ is split, then $Stab(\nabla)$ is at least as large as S^1. Precisely, we had the following lemma:

Lem If E is a $k \neq 0$ $SU(2)$ bundle, then $stab(\nabla)$ is larger than $\pm I$ IFF ∇ is split, in which case $Stab(\nabla) \cong S^1$

Towards proving this lemma, we discussed parallel transport and holonomy.

The general question is what are the possibilities of $Stab(\nabla)$?

A priori, we know that $Stab(\nabla) \subseteq G_1$, and this fact actually restricts what $stab(\nabla)$ must be as follows.

Lemma $g \in Stab(\nabla)$ IFF $D_\nabla g = 0$, where $D_\nabla : End(E) \to End(T^*M \otimes E)$

ie, g is parallel transported by D_∇ along every γ.

Pf $D_\nabla g$ is characterized by the Leibniz rule:

$$\nabla(gs) = (D_\nabla g)s + g(\nabla s)$$

After moving terms, we get

$$g^{-1}\nabla(gs) = g^{-1}(D_\nabla g)s + \nabla s$$

By definition, $g^*(\nabla)s = g^{-1}\nabla(gs)$

Thus $g \in Stab(\nabla)$ IFF $g^*(\nabla) = \nabla$

$$\Leftrightarrow \quad \nabla(s) = g^{-1}(D_\nabla g)s + \nabla s \qquad \forall s$$

$$\Leftrightarrow \quad g^{-1}(D_\nabla g)s = 0 \qquad \forall s$$

$$\Leftrightarrow \quad D_\nabla g \equiv 0 \qquad \square$$

Moreover, we know that the parallel transport is unique.

So we can also conclude the following.

Lemma If $g \in \text{Stab}(\nabla)$, then g is determined by its value at a single point.

Rmk Its not true that any g can be transported: g need not be globally defined! But we know $g \in \text{Stab}(\nabla)$, so a priori, it is a global object, so parallel transport makes sense.

Equivantly, we can write this as for any p, the restriction map $\mathcal{G} \to Aut(E_p)$ is injective on $\text{Stab}(\nabla)$.
$g \mapsto g_p$

Thus (via restriction) $\text{Stab}(\nabla)$ can be considered as a subgroup of $SU(2)$. In fact, $\text{Stab}(\nabla)$ has a nice connection with the holonomy of ∇.

Recall that for a fixed $p \in M$, the holonomy gives a map

$$hol_p : \{ \text{loops based at } p \} \to Aut(E_p) = SU(2)$$

$$\gamma \longmapsto hol_p^{\nabla}(\gamma)$$

Very explicitly, under hol_p, each γ gets sent to a specific linear transformation E_p to E_p, eg, a matrix transformation of the fibre, eg, an element of $SU(2)$. If γ is varied to another γ', we get another matrix transformation.

By considering every γ, the set of corresponding matrices is $\text{im}(hol_p)$, and since $hol_p(\gamma_2 \circ \gamma_1) = hol_p(\gamma_2) \circ hol_p(\gamma_1)$, $\text{im}(hol_p)$ forms a subgroup of $SU(2)$.

Lemma The restriction of $\mathrm{Stab}(\nabla)$ to a point p is the centralizer of $\mathrm{im}\,\mathrm{hol}_p$.

Pf ($\subseteq$): Take $g \in \mathrm{Stab}(\nabla)$.

We want to show that g_p commutes with $\mathrm{hol}_p(\gamma)$ for every loop γ.

We know that $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$.

Since $g \in \mathrm{Stab}(\nabla)$, we know $g^*(\nabla) = \nabla$, so $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = \mathrm{hol}_p^{\nabla}(\gamma)$.

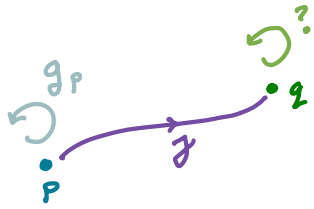
Thus $\mathrm{hol}_p^{\nabla}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$ as desired. □

($\supseteq$): Take $g_p \in \mathrm{Aut}(E_p)$. Further assume that $g_p \in Z(\mathrm{im}(\mathrm{hol}_p))$.

We want to show that there exists some $g \in \mathrm{Stab}(\nabla)$ such that the restriction of g to E_p is g_p.

Since $\mathrm{Stab}(\nabla) \equiv \{g \mid D_\nabla g = 0\}$, to define g, we should try to extend g_p via parallel transport by D_∇.

For any other point q, let γ be a path from p to q



Define g_γ such that

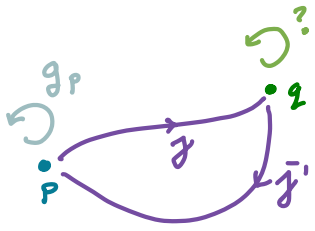
① g_p is extended along γ such that $(D_\nabla)_\gamma g = 0$

② $g_q = \mathrm{hol}_{p,q}(\nabla) \circ g_p \circ \mathrm{hol}_{q,p}(\bar{\gamma})$

We can do this along γ since parallel transport exists.

For any path, we can cook up a g_γ such that $D_\nabla(g_\gamma)$ vanishes along γ; this follows by existance of solutions to ODEs.

However, a priori, we don't know that this extension along γ gives a well-defined global g. Take γ' another path from p to q, and consider g' defined by γ'.



Then g_q and g'_q are defined as

$g_q = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma})$

$g'_q = \mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}')$.

If g_q is well defined, $g_q^{-1} = (g_q')^{-1}$ for any other g_q', i.e.,
$g_q (g_q')^{-1} = id$.

Indeed, $g_q (g_q')^{-1} = hol(\gamma) \circ g_p \circ hol(\tilde{\gamma}) \left(hol(\gamma') \circ g_p \circ hol(\tilde{\gamma}') \right)^{-1} = id$ □

Let's take stock so far: we know that $stab(\nabla)$ is determined by its restriction to a point which is the centralizer of the holonomy at that point, viewed as a subgroup of $SU(2)$.

So our question now becomes what are the subgroups of $SU(2)$?

Fact They are as follows:

Subgroups of $SU(2)$	Centralizer = $stab(\nabla)$
I	$SU(2)$
$-I$	$SU(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
$SU(2)$	$\pm I$

This means that the only possibilities for $stab(\nabla)$ are $\pm Id$, S^1, or $SU(2)$ itself. We are now in the position to prove the lemma.

Lemma If E is a $k \neq 0$ $SU(2)$ bundle, then $stab(\nabla)$ is larger than $\pm I$ IFF ∇ is split, in which case $stab(\nabla) \cong S^1$

Proof of lemma

We first show that either $stab(\nabla) \cong S^1$ or $\pm Id$. From there, we'll show the former happens IFF ∇ is split.

We just saw that there are only three possibilities for $stab(\nabla)$, so we need only to rule out $SU(2)$.

If ∇ is a connection on an $k \neq 0$ bundle, then F_∇ cannot

be identically 0 (if it was, $c_2 = k = 0$).

Thus holonomy can't be nontrivial, since the holonomy picks up curvature enclosed by a loop. If this path is varied, hol will vary continuously, so $\mathrm{im}(\mathrm{hol}_p)$ can't be contained in a discrete subgroup. Thus the centralizer cannot be $\pm I$, so the subgroup can't be $SU(2)$.

Now, suppose $\mathrm{stab}(\nabla) = S^1$. We just saw that $\mathrm{stab}(\nabla) = S^1$ implies that at a point $p \in M$, the element $g_p \in SU(2)$ must actually be contained in $U(1) \subset SU(2)$.

Therefore $g_p = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$, which induces a splitting of the fibre over p by considering the eigenspaces.

For any other $q \in M$, g_q is obtained from g_p via conjugation with $\mathrm{hol}_{p,q}(\nabla)$, so the eigenspaces of g_q provide a splitting of E_q compatible with that of E_p. Since $\mathrm{Stab}(\nabla)$ is reconstructed from g_p by parallel transport, the splitting of E_p induces a splitting of E.

To Do: Insert proof of other direction. □

Having proved this lemma, we can conclude our discussion of why the number of singular points in $\mathcal{M}$ is the number of splittings of E.

We'll now begin a discussion of the other major qualitative feature of $\mathcal{M}$: that $\partial\overline{\mathcal{M}} = M$. To do so, we'll first discuss instantons ((A)SD Yang-Mills connections) on S^4, which we'll do by identifying S^4 with $\mathbb{HP}^1$.

Interlude: Review of Quaternions & some useful isomorphisms from algebra

Def A quaternion is an expression of the form $q = x + yi + zj + wk$ where $x, y, z, w \in \mathbb{R}$ and $1, i, j, k$ are basis elements such that

$1 \cdot \alpha = \alpha$ for $\alpha = i, j, k$, $i^2 = j^2 = k^2 = ijk = -1$

$i \cdot j = k$, $j \cdot k = i$, $k \cdot i = j$, and $\alpha \cdot x = x \cdot \alpha$ for $\alpha = i, j, k$ and $x \in \mathbb{R}$

The space of quaternions is denoted $\mathbb{H}$

Remark $\mathbb{H}$ admits the structure of a division ring. However, multiplication fails to be commutative, so $\mathbb{H}$ is not a field: if $q_1 = i$ and $q_2 = j$, $q_1 q_2 = ij \neq ji = q_2 q_1$.

Note that $\mathbb{H} \cong \mathbb{C}^2 \cong \mathbb{R}^4$ by the following isomorphisms.

First rewrite a quaternion as

$$\begin{aligned} x + yi + zj + wk &= (x + yi) + (zj + wk) \\ &= (x + yi) + (zj + wij) \\ &= (x + yi) + (z + wi)j \end{aligned}$$

Then the isomorphisms are given by the maps

$$\begin{array}{ccccc} \mathbb{H} & \longrightarrow & \mathbb{C}^2 & \longrightarrow & \mathbb{R}^4 \\ (x+iy) + (z+iw)j & \mapsto & (x+iy, z+iw) & \mapsto & (x, y, z, w) \end{array}$$

The quaternions can also be identified by a specific subgroup of the space of 2×2 complex matrices.

Def The Pauli Matrices are a set of three 2×2 matrices $\{\sigma_1, \sigma_2, \sigma_3\} \subset Mat_{2\times 2}(\mathbb{C})$ which are traceless, Hermitian, unitary, and involutive:

$$\sigma_1 := \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \quad \sigma_2 := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \sigma_3 := \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

By sending $i \mapsto \sigma_1$, $j \mapsto \sigma_2$, and $k \mapsto \sigma_3$, any quaternion $q = x + yi + zj + wk$ can be rewritten as the linear combination

$$x\,Id + y\sigma_1 + z\sigma_2 + w\sigma_3 = \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} + \begin{bmatrix} yi & 0 \\ 0 & -yi \end{bmatrix} + \begin{bmatrix} 0 & z \\ -z & 0 \end{bmatrix} + \begin{bmatrix} 0 & wi \\ wi & 0 \end{bmatrix}$$

$$= \begin{bmatrix} x+yi & z+wi \\ -z+wi & x-yi \end{bmatrix}$$

Now, notice that the (2,2) entry is the complex conjugate of the (1,1) entry, and the (2,1) entry is the negative conjugate of the (1,2) entry. This relationship is clearly from the chosen basis vectors, so it holds for the image of any quaternion:

$$q \longmapsto \begin{bmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{bmatrix} \quad \text{where} \quad q = \alpha + \beta j$$

By now, matrices of this form should be familiar. If we divide by the norm, we recover the $SU(2)$ matrices!

Proposition The Lie group $SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is isomorphic (as a ring) to the ring of unit quaternions.

Recall that $SU(2)$ is the 2×2 complex matrices of determinant one. By identifying $\mathbb{C}^2$ with $\mathbb{R}^4$, an element of $SU(2)$ determines a determinant one 4×4 matrix, ie, a rotation in $\mathbb{R}^4$. Rotations in $\mathbb{R}^4$ are determined by a rotation of the unit sphere $S^3 \subset \mathbb{R}^4$, which is determined by the image of the $(1,0,0,0)$ vector, which is a point on S^3. This suggests the following claim.

Proposition $SU(2)$ is diffeomorphic to S^3

Exercise Prove this proposition

All together, we can conclude that $SU(2) \cong \{q \in \mathbb{H} \mid |q|^2 = 1\} \cong S^3$

We can now ask: what is $\mathbb{HP}^1$?

We now have the tools to understand why $\mathbb{HP}^2 \cong S^4$.

Recall that one construction of the projective spaces $\mathbb{RP}^1$ and $\mathbb{CP}^1$ are as the one point compactifications of $\mathbb{R}$ and $\mathbb{C}$, respectively; Since $\mathbb{H} \cong \mathbb{R}^4$ and $\mathbb{R}^4 \cup \{\infty\} \cong S^4$, we can conclude that $\mathbb{HP}^2 \cong S^4$!

Alternatively, recall that $\mathbb{RP}^n$ is most commonly constructed as the quotient of S^n by the antipodal map. By viewing the antipodal map as the action of the "unit circle" $\{\pm 1\} \subset \mathbb{R}$, and further identifying $\{\pm 1\} \cong S^0$, $\mathbb{RP}^n$ can be seen as the base space of a fibration $p: S^n \to \mathbb{RP}^n$ with S^0 fibres. This construction is easily generalized to $\mathbb{CP}^n$ and then $\mathbb{HP}^n$: Let V be a vector space over the ring $k = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$, and let $U(V^d)$ be the unit circle in V^d.

When $k = \mathbb{R}$, V^{n+1} is $n+1$ real dimensions, so $U(V_{\mathbb{R}}^{n+1}) \cong S^{n+1-1} = S^n$.

The unit circle in $\mathbb{R}$ is $\{\pm 1\} \cong S^0$. This recovers the fibration

$$S^0 \to S^n \to \mathbb{RP}^n.$$

When $k = \mathbb{C}$, V^{n+1} is $2(n+1)$ real dimensions, so $U(V_{\mathbb{C}}^{n+1}) \cong S^{2(n+1)-1} = S^{2n+1}$

The unit circle in $\mathbb{C} \cong \mathbb{R}^2$ is $\{e^{i\theta}\} \cong S^1$, so the fibration becomes

$S^1 \to S^{2n+1} \to \mathbb{CP}^n$. Note that when $n = 1$, this recovers the standard Hopf fibration $S^1 \to S^3 \to S^2$.

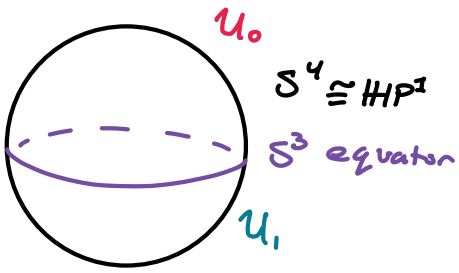
When $k \cong \mathbb{H}$, V^{n+1} is $4(n+1)$ real dimensions, so $U(V_{\mathbb{H}}^{n+1}) \cong S^{4(n+1)-1} = S^{4n+3}$

The unit circle in $\mathbb{H} \cong \mathbb{R}^4$ is S^3, thus the fibration is

$$S^3 \to S^{4n+3} \to \mathbb{HP}^1$$

Instantons on S^4

Mirroring our previous discussion of instantons on $\mathbb{CP}^1$, we'll view $S^4 = \mathbb{HP}^1$. $\mathbb{HP}^1$ admits two charts $U_0, U_1 \cong \mathbb{R}^4 \cong \mathbb{H}$, with the coordinate change function given by $q \longmapsto \frac{1}{q}$



What are $SU(2)$ bundles over S^4?

$SU(2)$ bundles correspond to maps $S^3 \to SU(2)$, where S^3 is the region on which we glue U_0 to U_1. Since $SU(2) \cong S^3$, $SU(2)$ bundles over S^4 are classified by maps $S^3 \to S^3$, of which there are $\mathbb{Z}$-many (degree)

In terms of coordinates, some possible gluing maps are $g(q) = q^n / |q|^n$

The $k=1$ bundle corresponds to $g(q) = q$.